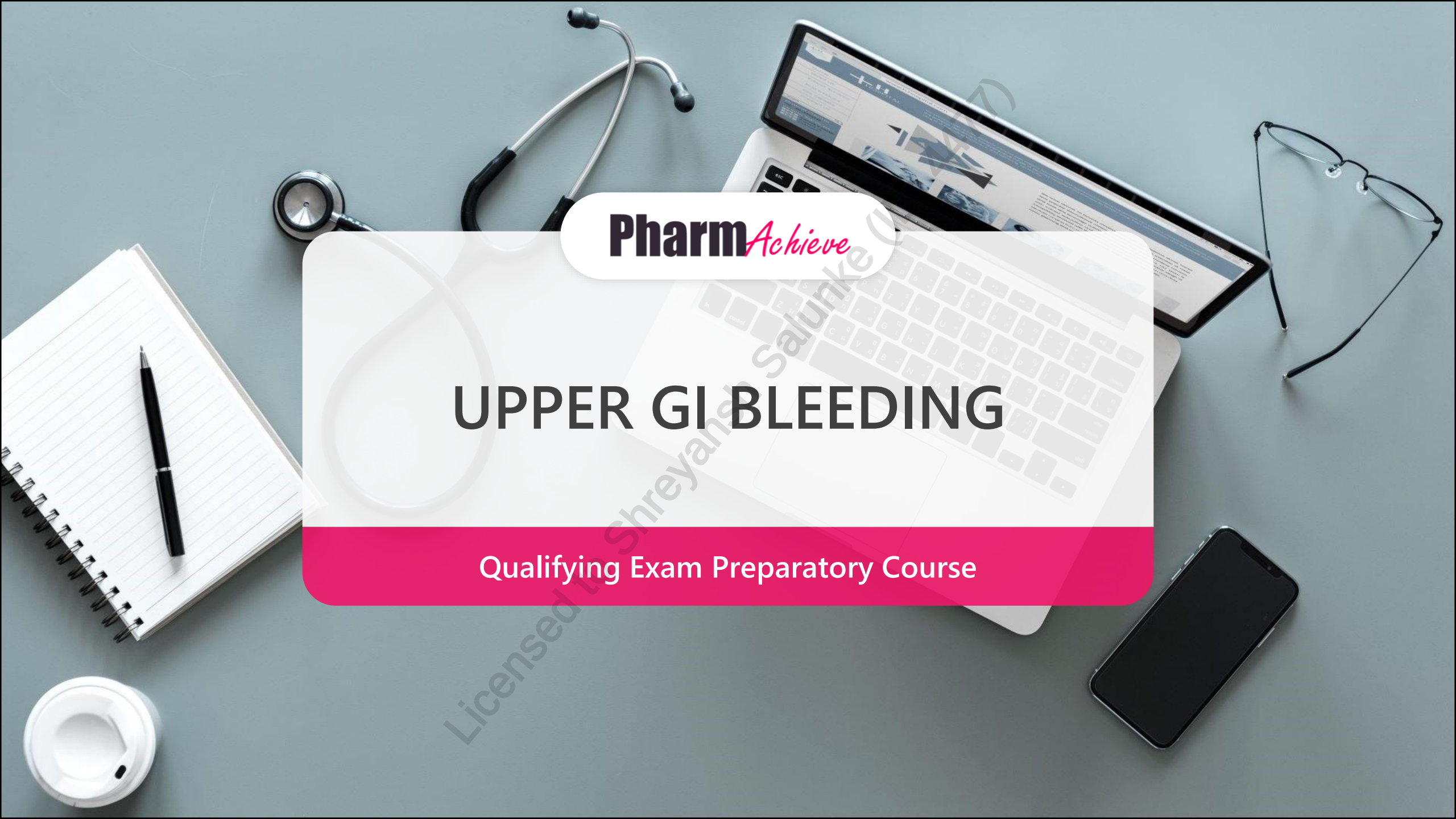




Pharm*Achieve*

UPPER GI BLEEDING

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

aPTT Activated Partial Thromboplastin Time

ASA Acetylsalicylic Acid

CBC Complete Blood Count

CDI Clostridium difficile Infection

EGD Esophagogastroduodenoscopy

GI Gastrointestinal

H2RA Histamine H2-Receptor Antagonist

INR International Normalized Ratio

IV Intravenous

NSAID Non-Steroidal Anti-Inflammatory Drug

NVNPUB Nonvariceal Nonpeptic Ulcer Bleeding

PPI Proton Pump Inhibitor

UGIB Upper Gastrointestinal Bleed

OBJECTIVES

- 1 Describe characteristics of upper gastrointestinal bleeding (UGIB)
- 2 Identify signs and symptoms of UGIB
- 3 Understand the importance of a risk assessment
- 4 Determine appropriate therapy

DEFINITION

Upper gastrointestinal bleeding (UGIB) refers to bleeding that starts in the esophagus, stomach or duodenum

- It is a **potentially fatal** condition with mortality rate of 3-14%
- Death mainly due to comorbid illnesses
- **80% spontaneously resolve without therapy**
- **Upper GI tract bleeding occurs approximately 4x more** than lower GI tract bleeding
- Greater incidence in **males** than females

Classified as:

- **Variceal bleeding**
 - Related to end-stage liver disease
- **Nonvariceal bleeding**
 - Other causes of UGIB (e.g. peptic ulcer disease)

CAUSES/RISK FACTORS

Variceal Bleeding

- **Esophageal Varices**
- Second most common cause of UGIB
- Generally occur at the **distal esophagus**
- **Hypertensive portal vein and hepatic cirrhosis obstructs venous blood flow and increases pressure**
- Severe rectal bleeding may occur by the same mechanism
- Risk factors: increased size of varices, severity of liver disease, prior variceal hemorrhages

CAUSES/RISK FACTORS

Nonvariceal Bleeding

Peptic Ulcer Disease (PUD)*

- Most common cause UGIB
- Duodenal ulcers, gastric ulcers
- PUD occurs mainly due to **H. Pylori infection or chronic NSAID use**

* Although PUD causes up to 50% of UGIB cases, reports of nonvariceal nonpeptic ulcer bleeding (NVNPUB) are rising significantly

Stress-Related Ulcers

- Common cause of acute UGIB
- Patients hospitalized and/or critically ill
- **High mortality rate**
- Risk factors: mechanical ventilation, coagulopathy, surgery, sepsis, burns, trauma

Mallory-Weiss Tears

- Longitudinal gastric mucosal tear that are **self-limited**
- Coughing, retching, and/or vomiting followed by hematemesis
- Relate to increase in intra-abdominal pressure
- Risk factors: excess alcohol use, diabetic ketoacidosis, hiatal hernias

Other causes of NVNPUB: malignancies, Dieulafoy's lesion, esophagitis, gastritis, other vascular lesions

CLINICAL PRESENTATION

GI Bleeding Presentation	
Hematemesis	• “Coffee-ground” emesis resulting from acid hemolysis • Indicates bleeding proximal to ligament of Treitz
Melena	• Black, tarry stool • Indicates proximal bleeding to the ligament of Treitz
Hematochezia	• Bright red or maroon coloured blood coming from the rectum
Occult GI bleeding	• Use of fecal occult blood test or iron deficiency, no obvious bleeding
Symptoms of hemorrhage	• Lightheadedness, tachycardia, angina, dyspnea, syncope

Signs and Symptoms Associated with Cause	
Variceal Hemorrhage	• Jaundice, weakness, fatigue, anorexia
PUD	• Abdominal pain, dyspepsia
Mallory-Weiss Tear	• Retching, coughing, vomiting prior to hematemesis

DIAGNOSIS

FYI

- Medical history
- Physical examination and vital signs
- Lab tests
 - CBC (serial)
 - Liver function tests
 - Coagulation tests (aPTT, PT/INR, anti-Xa)
 - Renal function
 - ECG (suspect ischemic heart disease)
- Risk assessment
 - Glasgow-Blatchford Score: use clinical and lab tests to identify low risk patients
 - Rockall Score: predict risk of rebleed and mortality with clinical and endoscopic information

GOALS OF THERAPY

- ✓ Minimize morbidity and mortality
- ✓ Identify the root cause of the hemorrhagic event
- ✓ Stabilize and treat the culprit lesion
- ✓ Prevent recurrences
- ✓ Prevent adverse effects from pharmacotherapy

NON-PHARMACOLOGICAL THERAPY

- Resuscitate and stabilize
 - Fluid resuscitation and perfusion management
 - Crystalloids (e.g. 0.9% sodium chloride, Ringer's lactate)
 - Vasopressors (with fluid/blood product replacement)
- Transfusion when hemoglobin < 80 g/L OR patient is unstable with frank hemorrhage
- Urgent endoscopy +/- hemostatic therapy if life-threatening bleeding
- Esophagogastroduodenoscopy (EGD)
 - Assess source of bleed
 - Assess risk of recurrent non-variceal bleed
- Endoscopic Hemostatic Intervention
 - Minimize surgery, rebleeding, mortality
 - Monotherapy or combination therapy



Protocols may differ depending on the hospital or institution

PHARMACOLOGICAL THERAPY: NONVARICEAL BLEEDING

- Proton Pump Inhibitors
- IV: Use pre-endoscopy and post-endoscopy in patients with high-risk stigmata
 - Pantoprazole: 80 mg bolus followed by continuous infusion 8 mg/h for 72 hours OR pantoprazole 40 mg IV BID (intermittent dosing)
 - Pre-endoscopic IV PPI not shown to reduce re-bleeding, surgery or mortality – but may have effect in potential high-risk patients (may reduce requirement for endoscopic therapy during index scope)
- Oral: Use for outpatients and post IV therapy inpatients
 - Lansoprazole, pantoprazole, omeprazole, esomeprazole, rabeprazole, dexlansoprazole

Tips:

- Reduces recurrent bleeding, risk of surgery compared to H2RA
- Oral or IV use of H2-receptor antagonists is not recommended
- Frequency and duration depends on the underlying cause
 - *H. pylori*: Continue usage until *H. pylori* successful eradication is confirmed
 - Ongoing ASA/NSAID use: may require extended duration for prevention



Protocols may differ depending on the hospital or institution

PHARMACOLOGICAL THERAPY: NONVARICEAL BLEEDING

- See lecture on peptic ulcer disease for detailed information

PHARMACOLOGICAL THERAPY: VARICEAL BLEEDING

- **Antibiotic prophylaxis:** Administer for 3-7 days
 - Oral **norfloxacin**
 - Oral or IV **ciprofloxacin**
 - IV 3rd generation **cephalosporin** (i.e. ceftriaxone)
 - **Preferred in advanced liver disease or areas with high fluoroquinolone resistance**

Tips:

- Start antibiotic therapy upon admission to hospital
 - ~ 20% are admitted with a bacterial infection
 - ~ 50% will develop peritonitis/bacterial infection
- **Antibiotic prophylaxis reduces risk of recurrent bleeding, bacterial infection, and mortality**

PHARMACOLOGICAL THERAPY: VARICEAL BLEEDING

- **Vasoactive drugs:** Administer for 3-5 days
 - **Octreotide**
 - **1st line therapy**
 - Somatostatin analogue
 - Reduce variceal blood flow
 - **Vasopressin & IV nitroglycerin** combination
 - **Not as effective or safe as octreotide**
 - Vasopressin: potent vasoconstrictor
 - IV nitroglycerin: reduces vasopressin side effects

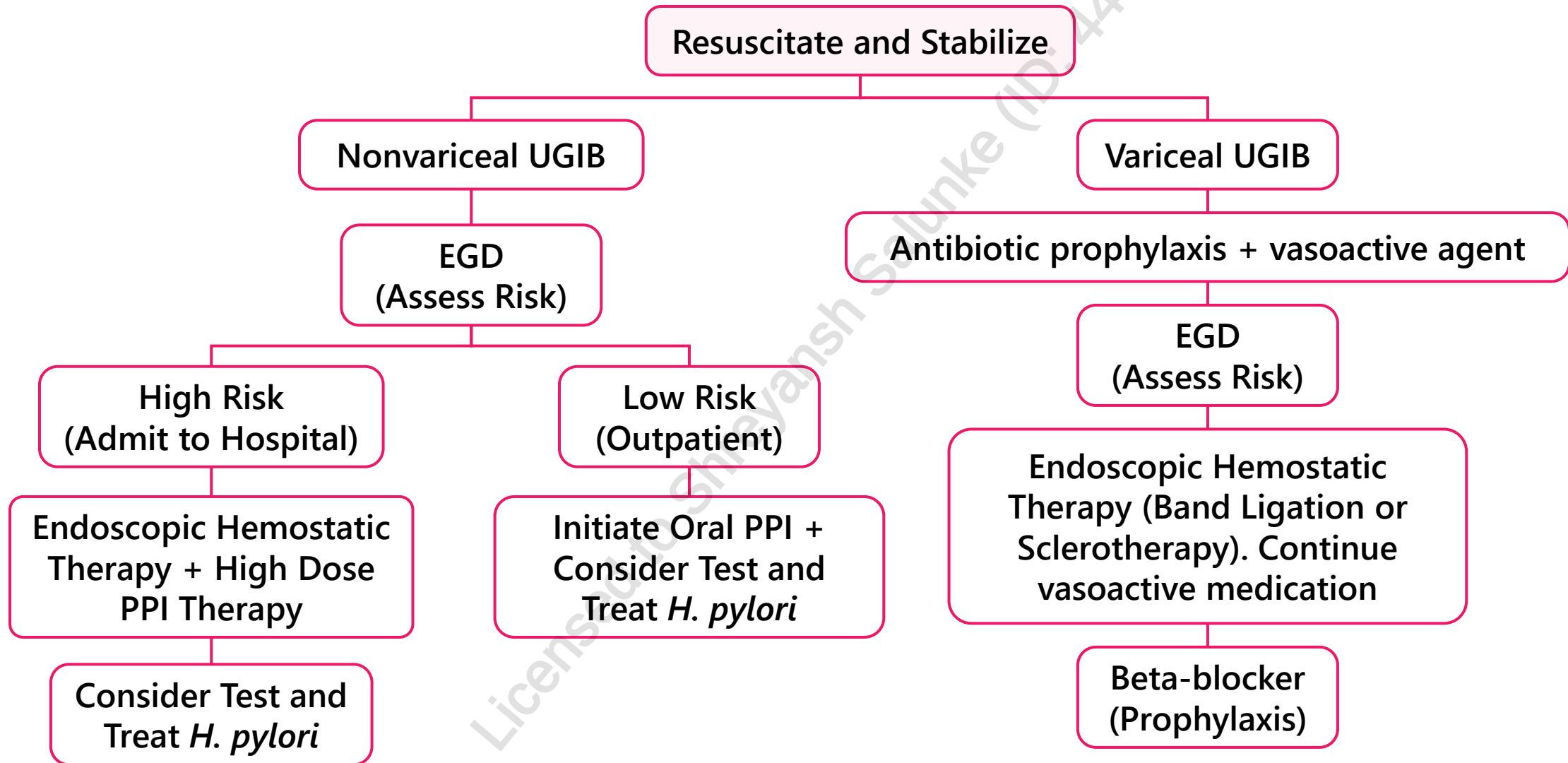
PHARMACOLOGICAL THERAPY: VARICEAL BLEEDING

- Non-selective β - blockers
 - Nadolol
 - Propranolol

Tips:

- Used as **prevention of variceal bleeding** in patients with portal hypertension and esophageal varices
- Monitor heart rate as a surrogate marker
- May be used as **monotherapy or in combination** with endoscopic variceal ligation to prevent recurrent bleeds

TREATMENT ALGORITHM



TAKEAWAY/THERAPEUTIC TIPS

- PPIs have greater efficacy than H2RAs for UGIB
- If endoscopic intervention is unavailable for the first 24 hours for a high-risk patient, then use pre-endoscopic IV PPI therapy
- Angiography and surgery may be used in aggressive or refractory cases
- Step down BID PPI regimens to once daily after 2 weeks of therapy and reassess total duration of PPI required
- Encourage avoidance of precipitants as possible
- Counsel on the complications of long-term PPI therapy (increased risk for CDI, pneumonia, fractures, hypomagnesemia, etc.)

CASE 1

HG is a 63-year-old female brought in by EMS with frank hematemesis and hypotension. Ms. HG has a history of alcohol use disorder. She was well until the morning of admission when she woke up and vomited large amounts of frank blood. She became lightheaded and dizzy and called 911. HG has a medical history of hepatitis B, cirrhosis, and alcohol use disorder. She currently takes nadolol 20 mg PO daily, furosemide 20 mg PO daily, and spironolactone 50 mg PO daily. She has no allergies to medications. She does not smoke and she drinks 2 bottles of wine per day.

Objective findings:

- BP=87/65 mmHg, HR=125 bpm
- Hgb=74, previous Hgb=95
- MCV elevated
- Plt=174
- AST/ALT/GGT elevated
- Albumin low



CASE 1

1. What type of bleed is she most likely having?
 - a) Non-variceal bleed
 - b) Variceal bleed
 - c) Peptic ulcer
 - d) Hemorrhoids

2. What therapeutic option would be most suitable for HG once variceal bleed is confirmed?
 - a) Ceftriaxone 1g IV q24h
 - b) Octreotide 50 mcg IV once, then 25 mcg/hour IV infusion
 - c) Pantoprazole 40 mg IV BID
 - d) Ceftriaxone and octreotide



CASE 1

3. HG undergoes endoscopic variceal ligation to stop the bleed. What would you recommend once she is stable 72 hours later?
- a) Pantoprazole 40 mg PO BID
 - b) Resume nadolol 20 mg PO daily
 - c) ciprofloxacin 500 mg PO BID
 - d) No medications are recommended at this time



CASE 2

GI is a 54-year-old male who presents to the ER with dizziness and pre-syncope, found to have a Hgb of 88. Mr. GI recently sprained his back after moving some furniture. He has been taking Motrin® (ibuprofen) up to 6 times a day to help with the pain at work as a construction worker. He noticed some stomach pain for the past few days, and yesterday noted that his stools were dark and tarry looking. GI has a medical history significant for hypertension, osteoarthritis, and GERD. He currently takes amlodipine 5 mg PO daily, Motrin® (ibuprofen) 400 mg PO Q4H PRN, Tylenol® Arthritis (acetaminophen) 650 mg PO QID PRN, vitamin D3 1000 units PO daily, and Gaviscon® (aluminum hydroxide and magnesium trisilicate) 1 tab PO QID PRN. He is allergic to ramipril resulting in angioedema. He smokes 10 cigarettes per day and drinks 1-2 beers a day.

Objective findings:

- BP=100/75 mmHg, HR=95 bpm
- Hgb=88, previous Hgb=123
- MCV normal
- Plt=260



CASE 2

1. What factor(s) increased GI's risk for a bleed?
 - a) Smoking
 - b) Osteoarthritis
 - c) Increased Motrin® use
 - d) GERD

2. What therapeutic option would be most suitable for GI?
 - a) Pantoprazole 40 mg PO daily and discharge
 - b) 1 unit packed red blood cell transfusion
 - c) Endoscopy +/- intervention
 - d) Famotidine 20 mg IV BID



CASE 2

3. What regimen would you recommend after endoscopy which showed a clean-based ulcer in the antrum and no intervention was undertaken?
- a) Lansoprazole 30 mg PO BID x 14 days, then once daily
 - b) Pantoprazole 80 mg IV bolus, then 8mg/h continuous infusion x 72h
 - c) Ranitidine 150 mg PO BID
 - d) Ceftriaxone 1g IV Q24H
4. All of the following are appropriate counselling points to GI upon discharge, EXCEPT?
- a) Avoid NSAID use
 - b) Limit alcohol use to no more than 4 standard drinks/day
 - c) Watch for symptoms of re-bleeding
 - d) Follow up and treat for *H.pylori* if positive



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CHANGE LOG

July 2025

- Slide 3: Legend updated
- Slide 5: Added definition of UGIB for clarity
- Slide 11: Updated hemoglobin transfusion value according to 2019 guideline
- Edited choices in case questions

November 2025

- Minor formatting edits, no content changes

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 7: Added note and other causes for NVNPUB
- Slide 17: Minor update in the algorithm of variceal UGIB